

Conjugation

Road map

Conjugation compares elements that differ only by a change of viewpoint inside the group. We first prove that conjugacy is an equivalence relation and compute all conjugacy classes of $S_3 \cong D_3$. We then introduce the centralizer and construct a bijection between the cosets of a centralizer and a conjugacy class. This gives the formula

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

Finally, we study the center $Z(G)$ and derive the class equation. All tables and partition diagrams in the handwritten lecture are reproduced and explained below.

5.1 Conjugation and conjugate elements

Definition 5.1: Conjugation by an element

Let G be a group and let $g \in G$. The map

$$\gamma_g : G \longrightarrow G, \quad a \longmapsto gag^{-1},$$

is called *conjugation by g* .

The element g acts on a from the left and g^{-1} acts from the right. Parentheses may be omitted because multiplication is associative. Conjugation is a bijection: its inverse is conjugation by g^{-1} , since

$$\gamma_{g^{-1}}(\gamma_g(a)) = g^{-1}(gag^{-1})g = a.$$

It also preserves multiplication:

$$\begin{aligned} \gamma_g(ab) &= gabg^{-1} \\ &= ga(g^{-1}g)bg^{-1} \\ &= (gag^{-1})(gbg^{-1}) = \gamma_g(a)\gamma_g(b). \end{aligned}$$

Thus conjugation is an automorphism of G , called an *inner automorphism*.

Definition 5.2: Conjugate elements

Elements $a, b \in G$ are *conjugate*, written

$$a \sim b,$$

if there exists $g \in G$ such that

$$b = gag^{-1}.$$

Proposition 5.1

Conjugacy is an equivalence relation on every group.

证明. We verify the three defining properties.

1. *Reflexivity*. Since $e^{-1} = e$,

$$a = eae^{-1},$$

so $a \sim a$.

2. *Symmetry*. If $a \sim b$, then $b = gag^{-1}$ for some $g \in G$. Multiplying by g^{-1} on the left and by g on the right gives

$$a = g^{-1}bg.$$

This is conjugation of b by g^{-1} , because $(g^{-1})^{-1} = g$. Hence $b \sim a$.

3. *Transitivity*. Suppose $a \sim b$ and $b \sim c$. Choose $g, g' \in G$ such that

$$b = gag^{-1}, \quad c = g'bg'^{-1}.$$

Substitution yields

$$\begin{aligned} c &= g'(gag^{-1})g'^{-1} \\ &= (g'g)a(g^{-1}g'^{-1}) \\ &= (g'g)a(g'g)^{-1}. \end{aligned}$$

Therefore $a \sim c$. ■

Definition 5.3: Conjugacy class

The *conjugacy class* of $a \in G$ is

$$\text{Cl}_G(a) := \{b \in G : b \sim a\} = \{gag^{-1} : g \in G\}.$$

When the ambient group is clear, we write simply $\text{Cl}(a)$.

Because conjugacy is an equivalence relation, the distinct conjugacy classes form a partition of G . In particular, every element belongs to exactly one conjugacy class.

Proposition 5.2: Basic invariants under conjugacy

If $b = gag^{-1}$, then a and b have the same order. More precisely, for every integer n ,

$$b^n = (gag^{-1})^n = ga^n g^{-1}.$$

证明. For $n \geq 1$, all adjacent factors $g^{-1}g$ cancel:

$$(gag^{-1})^n = ga \underbrace{(g^{-1}g)}_e a \cdots \underbrace{(g^{-1}g)}_e ag^{-1} = ga^n g^{-1}.$$

The formula for $n = 0$ is immediate, and negative powers follow by taking inverses. Consequently, $b^n = e$ if and only if $ga^n g^{-1} = e$, which is equivalent to $a^n = e$. Their orders are therefore equal. ■

5.2 Conjugacy classes in $S_3 \cong D_3$

Use the presentation

$$S_3 \cong D_3 = \langle a, b \mid a^3 = b^2 = e, ba = a^2b \rangle = \{e, a, a^2, b, ab, a^2b\}.$$

The complete Cayley table in the notes is

$\cdot$	e	a	a^2	b	ab	a^2b
e	e	a	a^2	b	ab	a^2b
a	a	a^2	e	ab	a^2b	b
a^2	a^2	e	a	a^2b	b	ab
b	b	a^2b	ab	e	a^2	a
ab	ab	b	a^2b	a	e	a^2
a^2b	a^2b	ab	b	a^2	a	e

The identity is fixed by every conjugation:

$$geg^{-1} = gg^{-1} = e,$$

so

$$\text{Cl}(e) = \{e\}.$$

Conjugating a by b gives

$$bab^{-1} = bab = (ba)b = (a^2b)b = a^2.$$

Since conjugate elements have the same order, a cannot be conjugate to e or to a reflection, all of which have order 2. Hence

$$\text{Cl}(a) = \text{Cl}(a^2) = \{a, a^2\}.$$

For the reflection b ,

$$\begin{aligned} aba^{-1} &= aba^2 = a(ba^2) = a(ab) = a^2b, \\ a^2ba^{-2} &= a^2ba = a^2(ba) = a^2(a^2b) = a^4b = ab. \end{aligned}$$

Therefore

$$\text{Cl}(b) = \text{Cl}(ab) = \text{Cl}(a^2b) = \{b, ab, a^2b\}.$$

The three distinct conjugacy classes are thus

$$\{e\}, \quad \{a, a^2\}, \quad \{b, ab, a^2b\}.$$

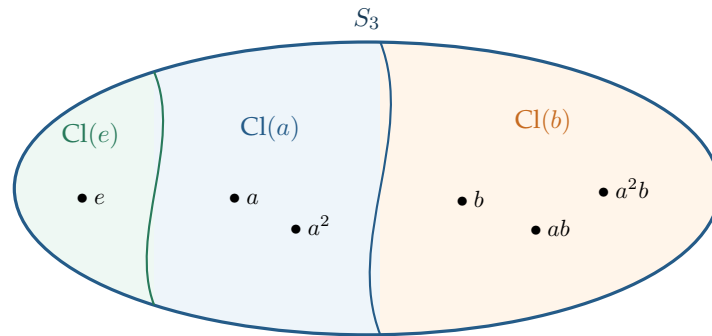


图 5.1: The partition of S_3 into its three conjugacy classes, of sizes 1, 2, and 3.

If G is Abelian, then

$$gag^{-1} = agg^{-1} = a$$

for every $a, g \in G$. Therefore every conjugacy class is a singleton:

$$\text{Cl}(a) = \{a\}.$$

The handwritten notes illustrate this with the Klein four-group

$$K_4 = \{e, a, b, c\},$$

whose four conjugacy classes are the four one-element sets.

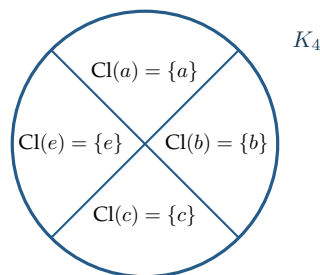


图 5.2: In the Abelian group K_4 , every conjugacy class is a singleton.

5.3 Centralizers

Definition 5.4: Centralizer of an element

For $a \in G$, the *centralizer* of a in G is

$$\text{Cen}_G(a) := \{b \in G : ab = ba\}.$$

It is the set of all elements of G that commute with a .

The defining condition has an equivalent conjugation form:

$$b \in \text{Cen}_G(a) \iff ba = ab \iff bab^{-1} = a.$$

Thus the centralizer is the set of group elements whose conjugation fixes a .

Proposition 5.3

For every $a \in G$, the centralizer $\text{Cen}_G(a)$ is a subgroup of G .

证明. The set is nonempty because $ea = a = ae$, so $e \in \text{Cen}_G(a)$. We use the one-step subgroup test. Let $x, y \in \text{Cen}_G(a)$. Then $ax = xa$ and $ay = ya$. The latter equality also implies $ay^{-1} = y^{-1}a$: from $ay = ya$, multiply by y^{-1} on the left and right to obtain $y^{-1}a = ay^{-1}$. Consequently,

$$\begin{aligned} a(xy^{-1}) &= (ax)y^{-1} \\ &= (xa)y^{-1} \\ &= x(ay^{-1}) \\ &= x(y^{-1}a) \\ &= (xy^{-1})a. \end{aligned}$$

Hence $xy^{-1} \in \text{Cen}_G(a)$, and the subgroup test gives $\text{Cen}_G(a) \leq G$. ■

If G is Abelian, every element commutes with a , so

$$\text{Cen}_G(a) = G \quad \text{for every } a \in G.$$

5.3.1 Centralizers in S_3

For $S_3 \cong D_3$, the centralizers recorded in the notes are

x	$\text{Cen}_{S_3}(x)$	$ \text{Cen}_{S_3}(x) $
e	S_3	6
a, a^2	$\{e, a, a^2\}$	3
b	$\{e, b\}$	2
ab	$\{e, ab\}$	2
a^2b	$\{e, a^2b\}$	2

For instance, an element x belongs to $\text{Cen}(a)$ exactly when the entry ax in the row of a agrees with the entry xa in the column of a . Reading those two lines of the Cayley table gives the comparison

x	ax	xa	$x \in \text{Cen}(a)?$
e	a	a	yes
a	a^2	a^2	yes
a^2	e	e	yes
b	ab	a^2b	no
ab	a^2b	b	no
a^2b	b	ab	no

In particular, the highlighted comparison in the handwritten table is

$$a(ab) = a^2b \neq (ab)a = b,$$

so $ab \notin \text{Cen}(a)$.

5.4 The size of a conjugacy class

The S_3 calculation suggests the identity

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

To prove it, we will show that the elements of $\text{Cl}_G(a)$ correspond one-to-one with the left cosets of $\text{Cen}_G(a)$.

Theorem 5.1: Centralizer–conjugacy-class formula

Let G be a group and $a \in G$. There is a bijection

$$G/\text{Cen}_G(a) \longrightarrow \text{Cl}_G(a), \quad g\text{Cen}_G(a) \longmapsto gag^{-1}.$$

If G is finite, then

$$|\text{Cl}_G(a)| = [G : \text{Cen}_G(a)]$$

and, equivalently,

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

证明. Write $C = \text{Cen}_G(a)$ and define

$$\Phi : G/C \longrightarrow \text{Cl}_G(a), \quad \Phi(gC) = gag^{-1}.$$

We verify all three requirements for a bijection.

Well-definedness. The same left coset may have different representatives. Suppose $gC = g'C$. The left-coset criterion gives

$$g^{-1}g' \in C.$$

Put $c = g^{-1}g'$, so $c \in C$ and $g' = gc$. Because c commutes with a , we have $cac^{-1} = a$. Therefore

$$\begin{aligned} g'ag'^{-1} &= (gc)a(gc)^{-1} \\ &= gcac^{-1}g^{-1} \\ &= gag^{-1}. \end{aligned}$$

Thus $\Phi(gC)$ does not depend on the chosen representative.

Surjectivity. If $b \in \text{Cl}_G(a)$, then by definition $b = gag^{-1}$ for some $g \in G$. Hence $b = \Phi(gC)$.

Injectivity. Suppose

$$\Phi(gC) = \Phi(g'C).$$

Then $gag^{-1} = g'ag'^{-1}$. Multiplying on the left by g^{-1} and on the right by g' yields

$$a(g^{-1}g') = (g^{-1}g')a.$$

Thus $g^{-1}g' \in C$, and the left-coset criterion gives $gC = g'C$. The map is therefore one-to-one.

We have proved that Φ is a bijection. When G is finite, Lagrange's theorem now gives

$$|\text{Cl}_G(a)| = |G/C| = [G : C] = \frac{|G|}{|C|}.$$

Multiplying by $|C| = |\text{Cen}_G(a)|$ produces the second formula. ■

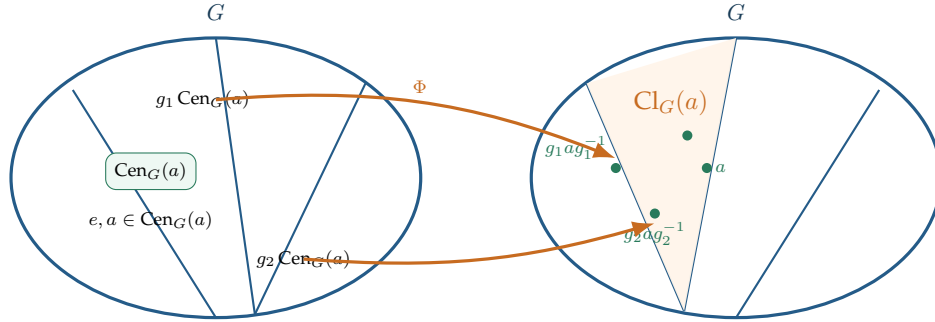


图 5.3: Each left coset of $\text{Cen}_G(a)$ determines exactly one conjugate of a . Different cosets determine different conjugates, which is precisely the bijection used in the theorem.

Two divisors of the group order

For a finite group, both $|\text{Cen}_G(a)|$ and $|\text{Cl}_G(a)|$ divide $|G|$: the first because the centralizer is a subgroup, and the second because it is the index of that subgroup.

Example 5.1: Checking the formula in S_3

Since $|S_3| = 6$,

representative x	$ \text{Cl}(x) $	$ \text{Cen}(x) $	$ \text{Cl}(x) \text{Cen}(x) $
e	1	6	$1 \cdot 6 = 6$
a	2	3	$2 \cdot 3 = 6$
b	3	2	$3 \cdot 2 = 6$

The notes explicitly record the first two lines:

$$|\text{Cl}(e)| = 1 = [S_3 : \text{Cen}(e)] = \frac{6}{6},$$

and

$$|\text{Cl}(a)| = 2 = [S_3 : \text{Cen}(a)] = \frac{6}{3}.$$

Proposition 5.4: Prime-order groups are Abelian

If G is finite and $|G| = p$ is prime, then G is Abelian.

证明. For every $a \in G$, the centralizer–class formula gives

$$|\text{Cl}(a)| |\text{Cen}(a)| = |G| = p.$$

If $a \neq e$, then the two distinct elements e and a both commute with a . Therefore

$$|\text{Cen}(a)| \geq 2,$$

so $|\text{Cen}(a)| \neq 1$. Since p is prime, the product formula forces

$$|\text{Cen}(a)| = p, \quad |\text{Cl}(a)| = 1.$$

The same singleton statement is obvious for $a = e$. Hence, for every $a, b \in G$,

$$bab^{-1} = a.$$

Multiplication on the right by b gives $ba = ab$. Thus every pair of elements commutes and G is Abelian. ■

There is also the Lagrange-theorem proof recorded in the notes. If $a \neq e$, then the cyclic subgroup

$$\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$$

contains at least e and a , so $|\langle a \rangle| \geq 2$. Its order divides the prime p , hence $|\langle a \rangle| = p$ and $\langle a \rangle = G$. Therefore every group of prime order is in fact cyclic, which also implies that it is Abelian.

5.5 The center of a group

Definition 5.5: Center

The *center* of G is

$$Z(G) := \{z \in G : az = za \text{ for every } a \in G\}.$$

Equivalently,

$$Z(G) = \bigcap_{a \in G} \text{Cen}_G(a).$$

The centralizer $\text{Cen}_G(a)$ contains the elements that commute with one fixed element a ; the center contains the elements that commute with *every* element of G . Consequently,

$$Z(G) \leq \text{Cen}_G(a) \leq G \quad \text{for every } a \in G.$$

The inclusions need not be proper.

Proposition 5.5: Elementary properties of the center

For every group G :

1. $Z(G) \leq G$;
2. $Z(G)$ is Abelian;
3. G is Abelian if and only if $Z(G) = G$;
4. $z \in Z(G)$ if and only if $\text{Cl}_G(z) = \{z\}$.

证明. First, $e \in Z(G)$. Let $x, y \in Z(G)$. For every $a \in G$,

$$\begin{aligned} a(xy^{-1}) &= (ax)y^{-1} \\ &= (xa)y^{-1} \\ &= x(ay^{-1}) \\ &= x(y^{-1}a) \\ &= (xy^{-1})a. \end{aligned}$$

Thus $xy^{-1} \in Z(G)$, and the subgroup test proves $Z(G) \leq G$.

If $x, y \in Z(G)$, then x commutes with every element of G , in particular with y . Hence $xy = yx$, so $Z(G)$ is Abelian.

If G is Abelian, every element commutes with every other element and therefore $Z(G) = G$. Conversely, if $Z(G) = G$, then every element of G commutes with every other element, so G is Abelian.

Finally, $z \in Z(G)$ means

$$gzg^{-1} = z \quad \text{for every } g \in G,$$

which is exactly the assertion that the conjugacy class of z contains only z . ■

A group is called *centerless* if

$$Z(G) = \{e\}.$$

For example, S_3 is centerless. Indeed, its three conjugacy classes are $\{e\}$, $\{a, a^2\}$, and $\{b, ab, a^2b\}$; only e lies in a singleton conjugacy class.

5.6 The class equation

Let G be finite, and choose one representative a_i from each conjugacy class. Since the conjugacy classes partition G ,

$$|G| = \sum_i |\text{Cl}_G(a_i)|.$$

The centralizer–class formula turns each summand into an index:

$$|\text{Cl}_G(a_i)| = [G : \text{Cen}_G(a_i)].$$

If $a_i \in Z(G)$, then its conjugacy class is the singleton $\{a_i\}$. Thus the central elements contribute exactly $|Z(G)|$ to the sum. Separating them from all nonsingleton conjugacy classes yields the principal counting formula.

Theorem 5.2: Class equation

Let G be a finite group. Choose one representative a_i from each conjugacy class not contained in $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : \text{Cen}_G(a_i)].$$

Equivalently,

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} |\text{Cl}_G(a_i)|.$$

证明. The distinct conjugacy classes form a disjoint partition of G , so their sizes add to $|G|$. A conjugacy class has one element precisely when its representative lies in $Z(G)$. There are $|Z(G)|$ such classes, each contributing 1. Every remaining class has a representative $a_i \notin Z(G)$ and has size $[G : \text{Cen}_G(a_i)]$. Adding these two kinds of contribution gives the displayed formula. ■

Example 5.2: The class equation of S_3

For S_3 ,

$$Z(S_3) = \{e\},$$

and representatives of the two noncentral conjugacy classes may be taken to be a and b . Therefore

$$\begin{aligned} |S_3| &= |Z(S_3)| + [S_3 : \text{Cen}(a)] + [S_3 : \text{Cen}(b)] \\ &= 1 + \frac{6}{3} + \frac{6}{2} \\ &= 1 + 2 + 3 = 6. \end{aligned}$$

Conjugacy class versus centralizer

The two sets associated with a answer complementary questions:

$$\text{Cl}_G(a) = \{bab^{-1} : b \in G\}$$

lists all elements obtained by moving a through conjugation, whereas

$$\text{Cen}_G(a) = \{b \in G : bab^{-1} = a\}$$

lists the group elements that leave a fixed. The larger the centralizer, the smaller the conjugacy class, exactly as expressed by

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

What to retain from this chapter

1. Conjugation is $\gamma_g(a) = gag^{-1}$, and conjugacy is an equivalence relation. Its equivalence classes partition G .
2. In $S_3 \cong D_3$, the classes are $\{e\}$, $\{a, a^2\}$, and $\{b, ab, a^2b\}$.
3. The centralizer $\text{Cen}_G(a) = \{b \in G : ab = ba\}$ is a subgroup.
4. Cosets of $\text{Cen}_G(a)$ correspond bijectively to conjugates of a , so

$$|\text{Cl}_G(a)| = [G : \text{Cen}_G(a)].$$

5. The center is $Z(G) = \bigcap_{a \in G} \text{Cen}_G(a)$, and central elements are precisely those whose conjugacy classes are singletons.
6. For a finite group,

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : \text{Cen}_G(a_i)].$$